

Representability criteria. Applications to subgroups of multiplicative type of affine group schemes

SGA 3, Exposé XI

English draft

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About this draft

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Contents

About this draft	iii
Exposé XI. Representability criteria. Applications to subgroups of multiplicative type of affine group schemes	1
0. Introduction	1
1. Review of smooth, étale, unramified morphisms	1
2. Examples of formally smooth functors drawn from the theory of groups of multiplicative type	6
3. Auxiliary representability results	8
4. The scheme of subgroups of multiplicative type of a smooth affine group	15
5. First corollaries of the representability theorem	19
6. On a rigidity property for homomorphisms of certain group schemes, and the representability of certain transporters	22

Exposé XI. Representability criteria. Applications to subgroups of multiplicative type of affine group schemes

by A. Grothendieck

0. Introduction

As we have already seen examples in Exp. X, Nos. 4, 5, the representability of certain functors, such as certain functors of the type $\underline{\mathrm{Hom}}_S(X, Y)$ and various variants, plays an important role in many questions concerning group preschemes.

Among the results particularly useful in this theory, let us mention (in addition to the questions of representability of quotients, studied in exposés V and VI and in Exp. VIII 5) the question of representability of functors of the form $\prod_{X/S} Y/X$ (Y a subobject of X), studied in Exp. VIII 6 in a very elementary case, of which variants will be given in no. 6 of the present exposé; these results provide us with the representability of various centralizers, normalizers, transporters.

Less elementary representability criteria, using results that will appear in EGA VI, are indicated in 6.12 and in Exp. XV, XVI, where a criterion for representability of quotients G/H in cases not covered by the preceding exposés will be given (a criterion that was not developed in the oral exposés).

Our main object in the present exposé is the proof of theorems 4.1 and 4.2, which provide a typical example of a nonprojective construction technique (close to that which will be developed in EGA VI). It has moreover become apparent, since the oral exposé and the writing of the present text, that the affine hypotheses made in 4.1 and 4.2 can be eliminated to a large extent (cf. XV), and that, on the other hand, for the essential part of the theory developed in the following exposé, one can dispense with 4.1 and 4.2. Finally, it might be interesting to prove the analogue of these results for a general reductive (for example semisimple) group prescheme instead of a group of multiplicative type, in which case 4.1 and 4.2 will doubtless be the key result for the proof.

1. Review of smooth, étale, unramified morphisms

The reader is referred to EGA IV §§ 17 & 18, and, pending its publication, to SGA 1 I, II, III (where, however, certain noetherian hypotheses, inconvenient in applications, should be replaced by hypotheses of finite presentation).

Definition 1.1. Let S be a prescheme, F a functor $(\mathrm{Sch})_S^\circ \rightarrow (\mathrm{Ens})$. One says that F is *formally smooth* (resp. *formally unramified*^{*}, resp. *formally étale*) if for every S -prescheme S' , affine (in the absolute sense), and every subscheme S'_0 of S' defined by a nilpotent ideal $\mathcal{J}$, the map

$$F(S') \longrightarrow F(S'_0)$$

is surjective (resp. injective, resp. bijective). A prescheme X over S is said to be *formally smooth over S* (resp. *formally unramified over S* , resp. *formally étale over S*) if the

⁰_{xy} version of 5/12/08.

^{*}One will now rather say “net” instead of “unramified”.

corresponding functor is formally smooth (resp. formally unramified, resp. formally étale); one says that X is *smooth* over S (resp. *unramified* over S , resp. *étale* over S) if it satisfies the preceding condition and if, in addition, X is locally of finite presentation over S .

The interest of these definitions for X lies in the fact that, on the one hand, they are expressed in a remarkably simple way in terms of the functor represented by X (and in practice, X is often given as the object over S representing an explicitly described functor), and that, on the other hand, they are also expressed by remarkable properties concerning the local structure of X , which we shall recall in the following statements (for the proof we refer to *loc. cit.*).

Proposition 1.2. *Let X be a prescheme locally of finite presentation over S . Then:*

- (i) *For X to be smooth over S , it is necessary and sufficient that X be flat over S , and that its geometric fibers $X \otimes_S \text{Spec}(\overline{\kappa(s)})$ be regular schemes. More generally, for X to be smooth over S in a neighborhood of the point $x \in X$ (one then says that X is smooth over S at x), it is necessary and sufficient that X be flat over S at x and X_s be smooth over $\kappa(s)$ at x , i.e. $X_s \otimes_{\kappa(s)} \overline{\kappa(s)}$ be regular at the points (or simply, one point) above x .*
- (ii) *Suppose S , hence X , locally noetherian, let $x \in X$ and $s \in S$ be its image in S ; then the smoothness of X over S at x can be recognized on the local homomorphism $A = \mathcal{O}_{S,s} \rightarrow B = \mathcal{O}_{X,x}$ of noetherian local rings (or even on the local homomorphism $\hat{A} \rightarrow \hat{B}$ of their completions), by the following characteristic property: B is flat over A , and $B \otimes_A k$ is geometrically regular over k (the residue field of A), i.e. for every finite extension k' of k , $(B \otimes_A k) \otimes_k k' = B \otimes_A k'$ is a regular semilocal ring. When the residue extension $k(B)/k(A)$ is trivial, these conditions are also equivalent to the following: $\hat{B}$ is isomorphic as an $\hat{A}$ -algebra to a formal power series algebra $\hat{A}[[t_1, \dots, t_n]]$.*

Thus, from the “formal” point of view, the structure of X over S is that of the *standard affine space* $S[t_1, \dots, t_n]$ over S .

Proposition 1.3. *Let X be a prescheme locally of finite presentation over S . Then:*

- (i) *The following conditions are equivalent:*
 - a) *X is unramified over S .*
 - b) *The diagonal morphism $X \rightarrow X \times_S X$ is an open immersion.*
 - c) *$\Omega_{X/S}^1 = 0$.*
 - d) *The geometric fibers of X/S are discrete and reduced, i.e. isomorphic to sums of copies of the base field.*
 - e) *For every $s \in S$, the fiber $X \otimes_S \text{Spec} \kappa(s) = X_s$ is unramified over $\kappa(s)$, or again is isomorphic to a sum of spectra of finite separable extensions of $\kappa(s)$.*
- (ii) *One has analogous conditions of a pointwise nature for X to be unramified over S at a given point x (i.e. in a neighborhood of the said point); for example, it is necessary and sufficient that $\mathcal{O}_{X_s,x}$ be a finite separable extension of $\kappa(s)$, which can also be expressed in terms of the local homomorphism $A = \mathcal{O}_{S,s} \rightarrow B = \mathcal{O}_{X,x}$ by the condition that $B \otimes_A k$ be a finite separable extension of k (the residue field of A), i.e. $\mathfrak{r}(A)B = \mathfrak{r}(B)$ and $k(B)$ is a finite separable extension of $k(A)$. When A , hence also B , is noetherian, and $k(A) \xrightarrow{\sim} k(B)$, this also means that $\hat{A} \rightarrow \hat{B}$ is surjective.*

Thus, from the “formal” point of view, saying that X is unramified over S means that X is essentially a *sub-prescheme* of S . Observe also that conditions d) and e) are expressed solely in terms of the fibers of X/S .

Proposition 1.4. *Let X be a prescheme locally of finite presentation over S . For X to be étale over S , it is necessary and sufficient that it be smooth over S and unramified over*

S (trivial by definition), which allows the criteria 1.2 and 1.3 to be applied. One finds in particular:

- (i) For X to be étale over S , it is necessary and sufficient that it be flat over S and unramified over S . An analogous local criterion for X to be étale over S at a given point x .
- (ii) Suppose in addition S , hence X , locally noetherian. Then the fact that X is étale over S at a point x can be recognized on the local homomorphism $A = \mathcal{O}_{S,s} \rightarrow B = \mathcal{O}_{X,s}$ (and even on the local homomorphism $\hat{A} \rightarrow \hat{B}$) by the following characteristic property: B is flat over A , and $B \otimes_A k$ is a finite separable extension of k (the residue field⁽¹⁾ of A). When $k(A) \xrightarrow{\sim} k(B)$, this condition simply means that $\hat{A} \rightarrow \hat{B}$ is an isomorphism.

Thus, from the “formal” point of view, saying that X is étale over S simply means that X is locally isomorphic to S .

Remark 1.5. When X is locally of finite presentation over S , and a point $x \in X$ is given, then the fact that X is smooth (resp. unramified, resp. étale) over S at x , i.e. near x , can still be recognized on the functor $X : (\text{Sch})^\circ_S \rightarrow (\text{Ens})$ by the following property: for every S' over S , S' the spectrum of a local ring, every subscheme S'_0 of S' defined by a nilpotent ideal, and every S -morphism $u_0 : S'_0 \rightarrow X$ taking the closed point s' of S' to x , there exists at least one (resp. at most one, resp. exactly one) S -morphism $u : S' \rightarrow X$ extending it. This statement shows in particular that in definition 1.1 one can restrict oneself to S' that are local schemes (provided the functor considered is representable by a prescheme locally of finite presentation over S). On the other hand, when S is locally noetherian, in the preceding pointwise criterion one can even restrict oneself to S' that are artinian local schemes, and one can therefore make the same restriction on S' in definition 1.1 (provided S is locally noetherian and the functor considered is represented by a prescheme locally of finite type over S).

Remark 1.6. Of course, given a morphism $f : X \rightarrow Y$ of preschemes, one will say that f is smooth (resp. ...) if f makes X a smooth Y -prescheme (resp. ...). When X and Y are S -preschemes and f an S -morphism of finite presentation, these properties of f are expressed immediately in terms of the morphism of functors $F, G : (\text{Sch})^\circ_S \rightarrow (\text{Ens})$ defined by f : f is smooth (resp. unramified, resp. étale) if for every S -prescheme S' , S' affine (in the absolute sense), and every subscheme S'_0 of S' defined by a nilpotent ideal $\mathcal{J}$, the map

$$F(S') \rightarrow F(S'_0) \times_{G(S'_0)} G(S')$$

deduced from the commutative square

$$\begin{array}{ccc} F(S') & \longrightarrow & G(S') \\ \downarrow & & \downarrow \\ F(S'_0) & \longrightarrow & G(S'_0) \end{array}$$

is surjective (resp. injective, resp. bijective), i.e. for every commutative square of morphisms of functors over S (where S', S'_0 are as above and $i : S'_0 \rightarrow S'$ is the canonical immersion):

$$(Q) \quad \begin{array}{ccc} S'_0 & \xrightarrow{i} & S' \\ u_0 \downarrow & & \downarrow v \\ F & \xrightarrow{f} & G \end{array}$$

⁽¹⁾**N.D.E.** “Residue extension” has been replaced by “residue field”.

there exists at least one (resp. at most one, resp. exactly one) morphism

$$\begin{array}{ccc} & & S' \\ & \swarrow u & \\ F & & \end{array}$$

making the two corresponding triangles commutative:

$$u_0 = ui \quad \text{and} \quad v = fu.$$

This property for a homomorphism of functors $f : F \rightarrow G$ over S still makes sense even if the functors considered are not representable; one will say (if it holds) that the homomorphism of functors $f : F \rightarrow G$ is formally smooth (resp. formally unramified, resp. formally étale). One will note that it depends only on the homomorphism of functors $(\text{Sch})^\circ \rightarrow (\text{Ens})$ defined by F, G (cf. I 1.4.1), and not on the structural morphisms $F \rightarrow S$ and $G \rightarrow S$. An equivalent way of expressing the preceding definition, rather easier to handle in applications, is the following: *the morphism of functors $F \rightarrow G$ over S is said to be formally smooth (resp. formally unramified, resp. formally étale) when, for every S' over S and every morphism $S' \rightarrow G$, the functor $F' = F \times_G S'$ over S' is formally smooth (resp. formally unramified, resp. formally étale).*

Remark 1.7. Under the conditions of 1.6, when a “point of F ” with values in a field k is given, i.e. an element x of $F(\text{Spec}(k))$ or, what amounts to the same thing, a morphism $\text{Spec}(k) \rightarrow F$, one likewise says that f is formally smooth (resp. formally unramified, resp. formally étale) *at x* if the condition of the preceding definition holds whenever S' is a local scheme and the morphism $S'_0 \rightarrow F$ in diagram (Q) above is “compatible with the marked points” in the following sense: if k' denotes the residue field of S' and S'_0 at their closed point, the diagram

$$\begin{array}{ccc} \text{Spec}(k') & & \text{Spec}(k) \\ j \downarrow & & \downarrow x \\ S'_0 & \xrightarrow{u_0} & F \end{array}$$

(where $j : \text{Spec}(k') \rightarrow S'_0$ denotes the canonical immersion) can be completed to a *commutative diagram*

$$\begin{array}{ccc} & \text{Spec}(k'') & \\ & \swarrow \quad \searrow & \\ \text{Spec}(k') & & \text{Spec}(k) \\ \downarrow & & \downarrow \\ S'_0 & \xrightarrow{\quad} & F, \end{array}$$

where k'' is the spectrum of a field. When F, G are representable by S -preschemes X, Y and the S -morphism $f : X \rightarrow Y$ is locally of finite presentation, this condition means precisely (by 1.5) that $f : X \rightarrow Y$ is smooth at the point of X that is the image of $\text{Spec}(k)$ under $x : \text{Spec}(k) \rightarrow X$.

Remark 1.8. When the condition of the preceding definition holds on restricting oneself to artinian local S' , one will say that $F \rightarrow G$ is *infinitesimally smooth* (resp. *infinitesimally unramified*, resp. *infinitesimally étale*) at x , and one says that $F \rightarrow G$ is infinitesimally smooth (resp. ...) if it is so at every point x , in other words if the condition considered in 1.6 holds whenever S' is artinian local. This variant of the preceding notions is technically useful, for it is often more convenient to check, being a weaker notion, while frequently

being sufficient (for example if $F \rightarrow G$ is a morphism locally of finite presentation, with G representable by a locally noetherian prescheme...) to imply the strong condition.

Smooth morphisms of preschemes behave remarkably simply with respect to differential calculus. We shall restrict ourselves here to recalling the following property:

Proposition 1.9. *Let $f : X \rightarrow S$ be a smooth morphism of preschemes. Then $\Omega_{X/S}^1$ is a locally free module of finite type on X , its rank at a point $x \in X$ is equal to the dimension of the fiber X_s (where $s = f(x)$) near the point x .*

One calls this dimension the *relative dimension* of X over S at x . One will note that it is zero (when f is smooth at x) if and only if f is étale at x . In practice, this dimension can also be calculated in terms of the functor F represented by X , in the following way. Let ξ be a point of F with values in a field k , “localized at x ”, i.e. a morphism $\text{Spec}(k) \rightarrow F = X$ whose image is x . Consider the algebra $D(k) = k[t]/(t^2)$ of dual numbers over k , considered as an S -prescheme, and consider the map

$$F(D(k)) \rightarrow F(k)$$

deduced from the augmentation $D(k) \rightarrow k$; finally, let $F(D(k), \xi)$ be the inverse image of $\xi \in F(k)$ under this map. Then this set is naturally endowed with a vector space structure over k (in fact, it is the vector space dual to $\Omega_{X/S}^1 \otimes_{\mathcal{O}_{X,x}} k$), whose dimension is the relative dimension of X over S at x .

To describe the vector space law on $F(D(k), \xi)$ explicitly, it is useful to introduce more generally, as in exposé II, for every vector space V over k , the algebra $D_k(V) = k + V$ (V an ideal of square zero), and to consider $F(D_k(V), \xi)$, the inverse image of ξ under $F(D_k(V)) \rightarrow F(k)$, as a covariant functor in V with values in (Ens). It then suffices that this functor commute with products of two factors (which means that F transforms certain amalgamated sums of a very particular type into fibered products, compare exposé II, a condition always satisfied if F is representable) to conclude that the $F(D_k(V), \xi)$ and in particular $F(D(k), \xi)$ are endowed with vector space structures over k . One can thus define the relative dimension of F over X at the “point” ξ under conditions substantially broader than the representability of F .

In the present exposé, the fact that certain functors which we shall describe explicitly are representable by smooth preschemes over S will serve us chiefly through the following result, which will be for us the technical intermediary for passing from constructions on the completion of the local ring A of a point s of a noetherian scheme S to a local ring A' over A étale over A , which will give in particular a means of passing from s to neighboring points:

Proposition 1.10. *Let $f : X \rightarrow S$ be a smooth morphism of preschemes, s a point of S , and x a point of X above s such that $\kappa(x)$ is a finite separable extension of $\kappa(s)$. Then there exists a subscheme S' of S , étale over S , passing through x . Thus one can find an étale morphism $S' \rightarrow S$, a point s' of S' above s with residue extension equal to $\kappa(x)/\kappa(s)$, and an S -morphism $S' \rightarrow X$ taking s' to x .*

To construct S' , one simply takes a system $g_1, \dots, g_n$ of sections of $\mathcal{O}_X$ on a neighborhood of x that induce at x a regular system of parameters of the local ring of the fiber X_s at x ; the subscheme S' defined by the g_i is then étale over S at x , and, provided one shrinks S' , it will therefore be étale over S .

We shall use 1.10 when $\kappa(x) = \kappa(s)$, i.e. x is rational over $\kappa(s)$, i.e. x can be considered as a section of X_s over $\text{Spec}(\kappa(s))$. Then 1.10 has the nature of a theorem on extending sections (after an étale extension of the base). It takes a particularly simple form in the following special case:

Corollary 1.11. *Under the conditions of 1.10, suppose that S is the spectrum of a henselian local ring, and that $\kappa(x) = \kappa(s)$. Then there exists a section of X over S passing through x (uniquely determined if X is even étale over S at x).*

Indeed, since S is henselian, it follows under the conditions of 1.10 that S' contains an open subscheme that is finite over S and whose fiber at s is reduced to x . Since it is étale over S , it follows that it is isomorphic to S , whence the conclusion.—One will note that when S is the spectrum of a complete local ring, 1.10 or 1.11 is more or less equivalent to the classical “Hensel’s lemma”, and it sometimes happens that it is referred to by this name.

2. Examples of formally smooth functors drawn from the theory of groups of multiplicative type

We shall interpret, in the language introduced in the preceding no. , the results stated in IX 3 concerning infinitesimal extensions of a homomorphism from a group of multiplicative type (consequences of the vanishing of the Hochschild cohomology of such a group, established in Exposé I).

Proposition 2.1. *Let S be a prescheme, H a group of multiplicative type over S , G an S -group prescheme smooth over S ; consider the functor over S*

$$M_H = \underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$$

(whose value at S' over S is the set of homomorphisms of S' -groups from $H_{S'}$ to $G_{S'}$). This functor is formally smooth over S .

Cf. IX 3.6. More generally:

Corollary 2.2. *Let S, G be as above, consider a homomorphism $u : H_1 \rightarrow H_2$ of group schemes of multiplicative type over S , whence, with the preceding notation, a morphism of functors over S :*

$$M_{H_2} \rightarrow M_{H_1} \quad (M_{H_i} = \underline{\mathrm{Hom}}_{S\text{-gr}}(H_i, G)),$$

given by $w \mapsto w \circ u$. This homomorphism is formally smooth.

Indeed, by definitions 1.6, this is equivalent to the following statement: when S is affine, S_0 a subscheme defined by a nilpotent ideal, let

$$v : H_1 \rightarrow G$$

be a homomorphism of S -groups, and

$$w_0 : (H_2)_{S_0} \rightarrow G_{S_0}$$

a homomorphism of S_0 -groups such that $w_0 u_{S_0} = v_{S_0}$; there then exists a homomorphism of S -groups

$$w : H_2 \rightarrow G$$

extending w_0 , and such that $wu = v$. To see this, one first extends w_0 to a homomorphism of S -groups $w' : H_2 \rightarrow G$, which is possible by 2.1; consider then $v' = w'u : H_1 \rightarrow G$, which satisfies $v'_{S_0} = v_{S_0}$ by the hypothesis on w_0 ; thus, by VIII 3.6, there exists an element g of $G(S)$, whose image in $G(S_0)$ is the identity element, and such that $v = \mathrm{int}(g)v'$, whence $v = (\mathrm{int}(g)w')u$, and it will therefore suffice to take $w = \mathrm{int}(g)w'$.

Corollary 2.3. *With the notation of 2.1, consider $M_H = M$ as a functor with group of operators G (G acting by $(v, g) \mapsto \mathrm{int}(g) \circ v$). Then the corresponding morphism*

$$R : G \times_S M \rightarrow M \times_S M$$

defined by $(g, v) \mapsto (\mathrm{int}(g) \circ v, v)$ is a formally smooth morphism.

By means of a base change $S' \rightarrow S$, this is equivalent to the following statement:

Corollary 2.4. *With the notation of 2.1, let $v_1, v_2 : H \rightarrow G$ be two morphisms of S -groups, let $\text{Transp}(v_1, v_2)$ be the subfunctor of G formed by the g such that $\text{int}(g) \circ v_1 = v_2$. Then this functor is formally smooth over S . In particular (if $v_1 = v_2 = v$) the functor $\text{Centr}(v)$, a subgroup of G formed by the h such that $\text{int}(h)v = v$, is formally smooth over S .*

(N.B. The pair (v_1, v_2) can be considered as a section over S of the second member in the morphism R of 2.3, and $\text{Transp}(v_1, v_2)$ as the inverse image functor of the said section under R). Statement 2.4 itself is equivalent to the following: when S is affine and S_0 is a subscheme of S defined by a nilpotent ideal, for every $g_0 \in G(S_0)$ such that $\text{int}(g_0)(v_1)_{S_0} = (v_2)_{S_0}$, g_0 extends to a $g \in G(S)$ such that $\text{int}(g)v_1 = v_2$. To prove it, one first extends g_0 to a section g' of G over S , which is possible since G is smooth over S ; one puts $v'_2 = \text{int}(g')v_1$, notes that v_2 and v'_2 have the same restriction over S_0 , so by the result IX 3.6 already invoked there exists a $g'' \in G(S)$ inducing the identity section on S_0 and such that $v_2 = \text{int}(g'')v'_2$, whence $v_2 = \text{int}(g'')\text{int}(g')v_1 = \text{int}(g''g')v_1$, so it suffices to take $g = g''g'$.

Proposition 2.1 bis. *Let S be a prescheme, G an S -group prescheme smooth over S ; consider the functor $M : (\text{Sch})^\circ_S \rightarrow (\text{Ens})$:*

$$M(S') = \text{set of subgroups of multiplicative type of } G_{S'}.$$

Then M is formally smooth over S .

Cf. IX 3.6 bis.

Corollary 2.2 bis. *Let n be an integer, and consider the morphism of functors*

$$\varphi_n : M \rightarrow M$$

defined by

$$\varphi_n(H) = {}_nH = \text{Ker}(n \cdot \text{id}_H).$$

Then φ_n is a formally smooth morphism. If for every integer $p > 0$, M_p denotes the subfunctor of M such that $M_p(S')$ is the set of subgroups H of multiplicative type of $G_{S'}$ such that ${}_pH = H$, then the morphism induced by φ_n :

$$M_{np} \rightarrow M_n$$

is formally smooth.

The second assertion is trivially contained in the first and is included only for the convenience of a later reference. The proof of the first is analogous to that of 2.2, invoking this time IX 3.6 bis.

Corollary 2.3 bis. *With the notation of 2.1 bis, consider M as a functor with group of operators G (G acting by $(g, H) \mapsto \text{int}(g)(H)$). Then the corresponding morphism*

$$R : G \times_S M \rightarrow M \times_S M$$

defined by $(g, v) \mapsto (\text{int}(g)v, v)$ is formally smooth.

This is equivalent to the following statement:

Corollary 2.4 bis. *With the notation of 2.1 bis, let H_1, H_2 be two subgroups of multiplicative type of G , and let $\text{Transp}_G(H_1, H_2)$ be the subfunctor of G formed by the g such that $\text{int}(g)H_1 = H_2$. Then this functor is formally smooth over S . In particular, if $H_1 = H_2 = H$, the subfunctor $\text{Norm}_G(H)$ of G , the normalizer of H , is formally smooth over S .*

The proof is analogous to that of 2.4, again invoking IX 3.6 bis.

Proposition 2.5. *Let S be a prescheme, G an S -group prescheme smooth over S , K a sub- S -group prescheme, smooth over S or of multiplicative type, H an S -group prescheme of multiplicative type, $u : H \rightarrow G$ a homomorphism of S -groups, and denote by $\text{Transp}_G(u, K)$*

the subfunctor of G whose value, at an S' over S , is formed by the $g \in G(S')$ such that $\text{int}(g)u_{S'} : H_{S'} \rightarrow G_{S'}$ factors through $K_{S'}$. Then this functor is formally smooth over S .

The proof is analogous to that of 2.2, invoking IX 3.6 and X 2.1 (the latter in the case where K is of multiplicative type).

3. Auxiliary representability results

Proposition 3.1. *Let $F \rightarrow S$ be a functor over the prescheme S . The following conditions are equivalent:*

- (i) F is representable, and $F \rightarrow S$ is an open immersion (one also says simply that $F \rightarrow S$ is an open immersion).
- (ii) F is a sheaf for the faithfully flat quasi-compact topology, “commutes with inductive limits of rings”, $F \rightarrow S$ is a monomorphism, and finally the following condition holds: for every local prescheme S' over S , with residue field k , and every S -morphism $\text{Spec}(k) = S'_0 \rightarrow F$, there exists an S -morphism $S' \rightarrow S$ extending it.
- (iii) (When S is locally noetherian). F is a sheaf for the faithfully flat quasi-compact topology, “commutes with inductive limits of rings”, “commutes with adic projective limits of artinian local rings”, $F \rightarrow S$ is a monomorphism, and is infinitesimally étale (cf. 1.8).

Let us first clarify two points of terminology.

Remark 3.2. One says that a functor⁽²⁾ F over S “commutes with inductive limits (understood: filtered) of rings” if for every filtered projective system $(S'_i)_{i \in I}$ of affine S -schemes over an affine open subset of S , with rings A'_i , the natural homomorphism

$$(*) \quad \varinjlim_i F(S'_i) \rightarrow F(S') \quad (\text{where } S' = \text{Spec } A', \ A' = \varinjlim_i A'_i)$$

is bijective. One will note that S' is nothing other than the projective limit of (S'_i) in the category of preschemes (and even of all ringed spaces), so the condition considered has the nature of a right exactness condition (commutation with certain inductive limits in $(\text{Sch})^\circ_S$), just like the condition of being a sheaf for some topology. One should take care that the condition considered is essentially *relative*, i.e. involves the morphism $F \rightarrow S$ and not only the functor $F : (\text{Sch})^\circ \rightarrow (\text{Ens})$; precisely, in $(*)$ $F(S'_i)$ and $F(S')$ denote $\text{Hom}_S(S'_i, F)$, $\text{Hom}_S(S', F)$. Thus, when F is representable, the condition considered means that F is locally of finite presentation over S . (And we have used several times, in the last two exposés, the fact that a functor represented by an S -prescheme locally of finite presentation commutes with inductive limits of rings).

Remark 3.3. One says that a functor⁽²⁾ F over S *commutes with adic projective limits of artinian local rings* if for every S' over S that is the spectrum of a complete noetherian local ring A' , putting $S'_n = \text{Spec}(A'/\text{rad}(A')^{n+1})$, the natural map

$$(xx) \quad F(S') \rightarrow \varprojlim_n F(S'_n)$$

is bijective. One will note that this condition, which has the nature of a left exactness condition, is *satisfied whenever F is representable*. One sees easily that, unlike the condition of commutation with inductive limits of rings, it is intrinsic to F as an element of $\text{Ob}(\widehat{(\text{Sch})})$, i.e. does not involve the morphism $F \rightarrow S$.

Remark 3.4. Let F be a functor over S that is a sheaf for the Zariski topology, or, as one also says, that is “of a local nature”. (For this it suffices that F be a sheaf for a finer topology, such as the faithfully flat quasi-compact topology). Let (S_i) be a covering of S by open

⁽²⁾**N.D.E.** Contravariant.

subsets; then one verifies easily (by a method of gluing pieces) that F is representable if and only if the $F_i = F \times_S S_i$ are, which allows one, for example, to reduce to the case where S is affine. Suppose that the functor F of a local nature commutes with inductive limits of rings. Then, for F to be representable, it is necessary and sufficient that its restriction to the category of preschemes locally of finite presentation over S be representable. The “necessary” was pointed out in 3.2, the “sufficient” amounts to this: if X is a prescheme locally of finite presentation over S and $X \rightarrow F$ a morphism such that, for every S' locally of finite presentation over S , the induced morphism

$$\mathrm{Hom}_S(S', X) \rightarrow \mathrm{Hom}_S(S', F)$$

is bijective, then $X \rightarrow F$ is an isomorphism. Now this follows easily from the fact that X and F are two functors of a local nature that commute with inductive limits of rings.

Let us now prove 3.1. The implications (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iii) are obvious; let us prove the converse implications.

One has (ii) $\Rightarrow$ (i). Indeed, let U be the set of $s \in S$ such that the canonical monomorphism $\mathrm{Spec}(\kappa(s)) \rightarrow S$ factors through F . By the last condition (ii), for every $s \in U$, the canonical monomorphism $\mathrm{Spec}(\mathcal{O}_{S,s}) \rightarrow U$ factors through F . Noting that $\mathcal{O}_{S,s}$ is the inductive limit of the rings of the affine neighborhoods of s , it follows from the fact that F commutes with inductive limits of rings that for every $s \in S$, there exists an open neighborhood U^s such that the canonical immersion $U^s \rightarrow S$ factors through F . This implies $U^s \subset U$, so U is open. Since $F \rightarrow S$ is a monomorphism, and F is of a local nature, the S -morphisms $U^s \rightarrow F$ glue in the $U^s \cap U^{s'}$ ($s, s' \in U$), hence come from an S -morphism $U \rightarrow F$. It remains to prove that this is an isomorphism, hence that every S -morphism $S' \rightarrow F$ factors uniquely through U (where S' is an S -prescheme). Since $F \rightarrow S$ and $U \rightarrow S$ is a monomorphism, it amounts to the same thing to say that the structural morphism $S' \rightarrow S$ factors through U , which reduces us to the case where S' is the spectrum of a field, hence reduced to a single point s' . Let s be the point of S below s' ; I say that the S -morphism $S' \rightarrow F$ factors through $\mathrm{Spec}(\kappa(s)) = S_0 \rightarrow F$ (which implies $s \in U$ and will prove what we want).

It amounts to the same thing, since $S' \rightarrow S_0$ is covering for fpqc and F is a sheaf for this topology, that the two composites

$$S'' = S' \times_{S_0} S' \rightrightarrows S' \rightarrow F$$

are the same, which follows from the fact that $F \rightarrow S$ is a monomorphism.

One has (iii) $\Rightarrow$ (ii) (when S is locally noetherian). It suffices to prove the last condition of (ii), and moreover (by the preceding proof) it suffices to do so when S' is of the form $\mathrm{Spec}(\mathcal{O}_{S,s})$, with $s \in S$. Let $A = \mathcal{O}_{S,s}$, $A_n = A/\mathfrak{m}^{n+1}$, $S_n = \mathrm{Spec}(A_n)$; it then follows from the hypothesis that $F \rightarrow S$ is infinitesimally smooth that the given morphism $S_0 \rightarrow F$ extends to morphisms $S_n \rightarrow F$. Since $F \rightarrow S$ is a monomorphism, one thus obtains an element of $\varprojlim_n F(S_n)$, and since F commutes with adic projective limits of artinian local rings, the $S_n \rightarrow F$ come from a morphism $\mathrm{Spec}(\hat{A}) = \hat{S}' \rightarrow F$. Since $F \rightarrow S$ is a monomorphism, F a sheaf for fpqc, and $\hat{S}' \rightarrow S'$ covering for the said topology, this morphism $\hat{S}' \rightarrow F$ factors through $S' \rightarrow F$, which completes the proof.

Proposition 3.5. *Let S be a locally noetherian prescheme, $F \rightarrow S$ a functor over S , $(X_i, u_i)_{i \in I}$ a family of S -morphisms $u_i : X_i \rightarrow F$, where the X_i are preschemes locally of finite type over S . Suppose the following conditions hold:*

- a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, commutes with adic projective limits of artinian local rings.*
- b) *The $u_i : X_i \rightarrow F$ are monomorphisms, and are infinitesimally étale (cf. 1.8).*
- c) *The family of u_i is “set-theoretically surjective”.*

Under these conditions, F is representable by a prescheme locally of finite type over S (and the u_i are open immersions, which make the family of X_i an open covering of F).

Remark 3.6. Proceeding as at the end of remark 1.7, one defines, for every functor $F : (\text{Sch})^\circ \rightarrow (\text{Ens})$, an “underlying set” $\text{ens}(F)$ as a quotient set of the set of points of F with values in fields (for the equivalence relation specified in 1.7). When F is representable by X , one indeed recovers the underlying set of X . Obviously $\text{ens}(F)$ depends functorially on F , so if $G \rightarrow F$ is a morphism of functors, one can say that this morphism is set-theoretically surjective if the induced map $\text{ens}(G) \rightarrow \text{ens}(F)$ is surjective. This therefore also means that every point of F with values in a field k “comes from” a point of G with values in a suitable extension of k . This definition extends at once to the case of a family of morphisms $G_i \rightarrow F$, which specifies the meaning of c).

Let us prove 3.5. For this, introduce, for $(i, j) \in I \times I$,

$$X_{i,j} = X_i \times_F X_j,$$

and consider the projections

$$v_{i,j} : X_{i,j} \rightarrow X_i \quad \text{and} \quad w_{i,j} : X_{i,j} \rightarrow X_j.$$

I say that these latter are representable by open immersions. To see this, one applies criterion 3.1 (iii): $X_{i,j}$ satisfies the three exactness conditions (being a sheaf, commuting with $\varinjlim$ of rings and with adic $\varprojlim$ of artinian local rings), for F, X_i, X_j satisfy them, and these conditions are stable under finite projective limits, in particular under fibered products; since $X_i \rightarrow F$ is a monomorphism, so is $v_{i,j} : X_{i,j} \rightarrow X_i$, which is deduced from it by the base change $X_j \rightarrow F$, and symmetrically $w_{j,i}$ is a monomorphism; finally the condition “infinitesimally étale” is also preserved by base change. This proves that one is under conditions 3.1 (iii).

We can now use the $X_i, X_{i,j}, v_{i,j}$ and $w_{i,j}$ to construct in the usual way an S -prescheme X , such that the X_i are identified with open subsets of X , the $X_{i,j}$ with the intersections $X_i \cap X_j$ and the $v_{i,j}, w_{i,j}$ with the canonical immersions. Note that X is also the quotient of

$$\mathbf{X} = \coprod_i X_i$$

by the equivalence relation $R = \coprod_{i,j} X_{i,j}$ (the two projections $v, w : R \rightrightarrows Y$ being defined by the $v_{i,j}$, resp. the $w_{i,j}$). More precisely, since F is a sheaf for fpqc, the $u_i : X_i \rightarrow F$ come from a $u : \mathbf{X} \rightarrow F$, and R is nothing other than the equivalence relation defined by u , $R = \mathbf{X} \times_F \mathbf{X}$; finally the quotient $X = Y/R$ is also a quotient in the category of sheaves for fpqc (and even in the category of sheaves for the Zariski topology): it suffices to use the definitions of “quotient” and “sheaf” to convince oneself of this. Consequently u factors uniquely through a morphism

$$u : X \rightarrow F,$$

and this morphism is a monomorphism. It remains to show that it is an isomorphism. Since F is of a local nature, one can suppose S affine, and since moreover F commutes with inductive limits of rings, it suffices to verify that for every T affine of finite type over S , every morphism $T \rightarrow F$ factors through X (cf. 3.4). For this, consider $G = X \times_F T \rightarrow T$; it suffices to prove that this is an isomorphism. Now, since T is noetherian, one sees as above that it is an open immersion (N.B. $X \rightarrow F$ is infinitesimally étale, as follows at once from the fact that the induced morphisms $u_i : X_i \rightarrow F$ are). But by hypothesis $X \rightarrow F$ is set-theoretically surjective, and one sees immediately that this is a condition stable under base change, so $G \rightarrow T$ is set-theoretically surjective, hence an isomorphism since it is an open immersion. Q.E.D.

Proposition 3.7. *Let S be a locally noetherian prescheme, I an increasingly filtered index set, $(T_i)_{i \in I}$ a projective system of S -preschemes locally of finite type, $T = \varprojlim T_i$ the projective limit functor, F a functor over S , $u : F \rightarrow T$ an S -morphism. Suppose the following conditions hold:*

- a) *F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, and with adic projective limits of artinian local rings.*

- b) The morphism $u : F \longrightarrow T$ is a monomorphism.
- b') The morphism $u : F \longrightarrow T$ is infinitesimally étale.
- c) For every point ξ of F with values in the spectrum of a field k , denoting by $\xi_i \in T_i(\text{Spec}(k))$ its image and by t_i the corresponding element of T_i , there exists an $i \in I$ such that for $j \geq i$ the transition morphism $p_{i,j} : T_j \longrightarrow T_i$ is étale at t_j .
- d) For every prescheme X locally of finite type over S , and every S -morphism $X \longrightarrow F$, the set of $x \in X$ at which this morphism is infinitesimally étale is open.

Under these conditions, F is representable by a prescheme locally of finite type over S .

Let us note at once that in the case which will concern us in the following no. , conditions c) and d) will be verified by means of the following corollary:

Corollary 3.8. *Assuming a), b), b'), conditions c) and d) are implied by the following:*

- c') The T_i are smooth over S , and the transition morphisms $p_{i,j} : T_j \longrightarrow T_i$ are smooth.
- d') For every point ξ of F with values in the spectrum of a field k , let $t_i(\xi)$ be the element of T_i defined by ξ , $d_i(\xi)$ the relative dimension of T_i over S at $t_i(\xi)$, and $d(\xi) = \sup d_i(\xi)$. Then:

- 1°) for every ξ as above, one has $d(\xi) < +\infty$, and
- 2°) for every prescheme X locally of finite type over S , and every S -monomorphism $v : X \longrightarrow F$, the function $x \longmapsto d(\xi_x)$ on X is locally constant (where, for $x \in X$, ξ_x denotes the point of F with values in $\kappa(x)$ induced by v).

Let us prove 3.7. Place ourselves under the conditions of c), let t be the element of $\text{ens}(F)$ defined by ξ (cf. 3.6) and put

$$\mathcal{O}_t = \varinjlim_i \mathcal{O}_{T_i, t_i}.$$

Then, using the condition stated in c), one sees easily that $\mathcal{O}_t$ is a *noetherian* local ring (EGA 0_{IV} 10.3.1.3). Its residue field $\kappa(t)$ is the inductive limit of the residue fields $\kappa(t_i)$, and k is an extension of it. If η denotes the spectrum of $\kappa(t)$, ξ that of k , one has a commutative diagram

$$\begin{array}{ccc} \xi & \longrightarrow & F \\ \downarrow & & \downarrow \\ \eta & \longrightarrow & T, \end{array}$$

whose definition is obvious and left to the reader. Since $\xi \longrightarrow \eta$ is covering for the fpqc topology, F is a sheaf for that topology by a), and $F \longrightarrow T$ a monomorphism by b), one sees at once that the morphism $\eta \longrightarrow T$ factors (uniquely) through a morphism $\eta \longrightarrow F$. Let us now note the

Lemma 3.9. *Under the conditions of 3.7, let $T' = \text{Spec}(A)$ be a noetherian local scheme, $T'_0 = \text{Spec}(k(A))$, $i : T'_0 \longrightarrow T'$ the canonical immersion, and suppose given a commutative square of morphisms*

$$\begin{array}{ccc} T'_0 & \longrightarrow & F \\ i \downarrow & & \downarrow \\ T' & \longrightarrow & T. \end{array}$$

Then there exists a unique T -morphism $T' \longrightarrow F$.

The proof is that of 3.1 (iii) $\Rightarrow$ (ii) (where the fact that the S of the cited statement is representable was not used), using that F is a sheaf for the fpqc topology, commutes with adic projective limits of artinian local rings (in this case the $A/\text{rad}(A)^{n+1}$), and that $F \longrightarrow T$ is a monomorphism and is infinitesimally étale.

We shall apply lemma 3.9 to the case where $T' = \text{Spec}(\mathcal{O}_t)$, hence $T'_0 = \text{Spec}(\kappa(t))$, noting that we have just constructed $T'_0 \longrightarrow F$, and that the canonical homomorphisms

$\mathcal{O}_{T_i, t_i} \longrightarrow \mathcal{O}_t$ define a projective system of morphisms $\mathrm{Spec}(\mathcal{O}_t) \longrightarrow T_i$, whence a canonical morphism $\mathrm{Spec}(\mathcal{O}_t) \longrightarrow T$. The commutativity of the corresponding square (x) is trivial (for, by definition of T , it suffices to verify it with T replaced by T_i), whence a unique T -morphism

$$v' : T' = \mathrm{Spec}(\mathcal{O}_t) \longrightarrow F.$$

Since F commutes with inductive limits of rings, this morphism factors through

$$v'_i : T'_i = \mathrm{Spec}(\mathcal{O}_{T_i, t_i}) \longrightarrow F$$

for large i . For such an i , one has

$$T' \xrightarrow{\sim} T'_i \quad \text{i.e.} \quad \mathcal{O}_{T_i, t_i} \xrightarrow{\sim} \mathcal{O}_t.$$

Indeed, since $T' \longrightarrow T'_i$ is faithfully flat quasi-compact and hence an effective epimorphism, it suffices to see that it is a monomorphism. Now if one has two morphisms $R \rightrightarrows T'$ (with R representable) having the same composite with $T' \longrightarrow T'_i$, they have the same composite with $T' \longrightarrow T$ (which factors through $T' \longrightarrow T'_i$), hence the same composite with $T' \longrightarrow T_j$ for every j , hence with $T' \longrightarrow T'_j$ for every j , and are therefore equal since T' is the projective limit of the T'_j in the category (Sch).

Consider the diagram

$$\begin{array}{ccc} T' & \xrightarrow{v'} & F \\ \downarrow & \nearrow v'_i & \downarrow \\ T'_i & \longrightarrow & T_i, \end{array}$$

where the unspecified arrows are the obvious arrows. The square is commutative and the upper triangle also, so since $T' \longrightarrow T'_i$ is an epimorphism, it follows that the lower triangle is commutative. Now $\mathcal{O}_{T_i, t_i}$ is the inductive limit of the affine rings of the affine open neighborhoods of t_i in T_i , so since F commutes with inductive limits of rings, v'_i comes from a morphism

$$v_t : U_t \longrightarrow F$$

from an open neighborhood U_t of t_i in T_i . One sees immediately that, on restricting this neighborhood if necessary, v_t is necessarily a T_i -morphism (T_i being locally of finite type over S , hence also commuting with inductive limits of rings). Then the composite of v_t with $F \longrightarrow T_i$ is a monomorphism, so v_t is a monomorphism. I say that it is *infinitesimally étale at t_i* . Since $F \longrightarrow T$ is infinitesimally étale, it amounts to the same thing to say that the composite $T_i \longrightarrow F \longrightarrow T$ is infinitesimally étale at t_i , or again that the induced morphism $T'_i \longrightarrow T$ is infinitesimally étale, or finally (since $T' \xrightarrow{\sim} T'_i$) that $T' \longrightarrow T$ is infinitesimally étale, which is immediate, this morphism being the projective limit of the infinitesimally étale morphisms $T'_i \longrightarrow T_i$.

Let us now apply condition d) (which had not yet been used); it follows that v_t is infinitesimally étale near t , so on replacing U_t by a smaller open subset one can suppose that v_t is an infinitesimally étale monomorphism.

For variable $t \in \mathrm{ens}(F)$, the family of morphisms v_t falls under 3.5, which implies the conclusion of 3.7.

Let us now prove 3.8, supposing that conditions c') and d') of 3.8 hold. Then, with the notation of d'), one will have $d_i(\xi) = \text{constant}$ for large i , so the relative dimension of T_j over T_i at t_j , equal to $d_j(\xi) - d_i(\xi)$, is zero, hence $T_j \longrightarrow T_i$ is étale at t_j , which proves condition c). Consequently the preceding proof applies to give, for every $t \in \mathrm{ens}(F)$, an index i , an open neighborhood U_t of t_i and a morphism $U_t \longrightarrow F$ that is a monomorphism, infinitesimally étale at t_i , and everything amounts to proving that this morphism is infinitesimally étale near t_i . Now with the notation of d') 2°, where one takes $X = U_t$, one can suppose, on replacing U_t by the connected component of t_i in U_t , that for every $x \in U_t$ one has

$$d(\xi_x) = d(\xi_{t_i}) \quad \text{for every } x \in U_t.$$

But $d(\xi_{t_i}) = \text{relative dimension of } T_i \text{ over } S \text{ at } t_i$, and since the relative dimension of T_i over S remains constant on U_t , one will also have, for every $x \in U_t$:

$$(*) \quad d(\xi_x) = \text{relative dimension of } U_t \text{ over } S \text{ at } x.$$

On the other hand, with the notation of the proof of 3.7, one sees at once that for every noetherian local prescheme $R' = \text{Spec}(A)$, putting $R'_0 = \text{Spec } k(A)$, every morphism $R' \rightarrow F$ such that the induced morphism $R'_0 \rightarrow F$ factors through F (i.e. taking the closed point of R' to $t \in \text{ens}(F)$) factors (obviously uniquely) through T' . (The proof is that of 3.1 or 3.9: one uses that $T' \rightarrow F$ is a monomorphism infinitesimally étale at t , and that T' is a sheaf for fpqc commuting with $\varprojlim$...). Applying this result to the morphism $\text{Spec}(\mathcal{O}_{U_t, x}) = R' \rightarrow F$ induced by v_t and to the point $t_x \in \text{ens}(F)$ that is the image of the closed point of R' , i.e. the image of x under v_t , one finds a factorization

$$\text{Spec}(\mathcal{O}_{U_t, x}) \rightarrow \text{Spec}(\mathcal{O}_{t_x}) \rightarrow F$$

and since the second arrow is infinitesimally étale at t_x , to prove that the composite is so at x , it suffices to prove that the first is so at t_x .

Now, thanks to formula (x) above, it is a local S -homomorphism of the localizations, at two points x, t_x , of smooth S -preschemes U_t, U_{t_x} of the same relative dimension d over S at t, t_x . This morphism is induced by an S -morphism

$$w : U \rightarrow V$$

where U is an open neighborhood of x in U_t , and where $V = U_{t_x}$ (EGA I 6.5.1). Everything amounts to proving that this morphism is étale at x . Moreover U and V are endowed with monomorphisms $U \rightarrow F, V \rightarrow F$, and one sees at once that, on restricting U further, w is an F -morphism, so w is a monomorphism. It now suffices to prove the

Lemma 3.10. *Let U, V be two smooth S -preschemes, of the same relative dimension d over S , and $w : U \rightarrow V$ an S -morphism that is a monomorphism; then w is an open immersion (and a fortiori is étale).*

By SGA I 5.7 one is reduced to the case where S is the spectrum of a field, which one can suppose algebraically closed. By SGA I 5.1 it suffices to prove that w is étale, and it suffices to prove it at the closed points of U . Let x be such a point, $y = w(x)$; then, taking a regular system of parameters $f_1, \dots, f_d$ of $\mathcal{O}_{V, y}$, one sees that $A = \mathcal{O}_{U, x} / \sum f_i \mathcal{O}_{U, x}$ is the trivial extension of $k = \mathcal{O}_{V, y} / \sum f_i \mathcal{O}_{V, y}$ (for, since w is a monomorphism, so is the structural morphism $\text{Spec}(A) \rightarrow \text{Spec}(k)$ deduced from it by base change). Since $\mathcal{O}_{U, x}$ is a regular local ring of dimension d , it follows that the f_i form a regular system of parameters of this ring, which implies at once that w is étale at x and completes the proof of 3.10, hence of 3.8.

Corollary 3.11. *Under the conditions of 3.8, for every quasi-compact open subset U of F separated over S , there exists an $i \in I$ such that for every $j \geq i$ the morphism $u_j|_U : U \rightarrow T_j$ is an open immersion. In particular, if the T_i are quasi-affine over S , then every open subset U of F quasi-compact over S , i.e. of finite type over S , is quasi-affine over S .*

The proof of 3.7 shows that for every $t \in F$, there exists an $i \in I$ such that $u_i : F \rightarrow T_i$ is a local isomorphism at t , and then u_j is a local isomorphism at x for every $j \geq i$. By quasi-compactness, one can choose i independent of $x \in U$. It remains to prove that for large i , $u_i|_U : U \rightarrow T_i$ is a monomorphism. Now, since $U \rightarrow T$ is a monomorphism, one sees that the intersection of the equivalence relations $U \times_{T_i} U \subset U \times_S U$ is reduced to the diagonal, and since $U \times_S U$ is a noetherian prescheme and the $U \times_{T_i} U$ closed sub-preschemes, it follows that one of these $U \times_{T_i} U$ is already reduced to the diagonal, i.e. $u_i|_U$ is a monomorphism. This proves the first assertion in 3.11, and the second is an immediate consequence of it.

Proposition 3.12. *Let S be a prescheme, G an affine S -group prescheme.*

a) Let F be the functor $(\text{Sch})/S \longrightarrow (\text{Ens})$ such that, for every T over S ,

$$F(T) = \text{set of subgroups of multiplicative type of } G_{T_S} \\ \text{that are finite over } T.$$

Suppose S locally noetherian or G of finite presentation over S . Then the functor F is representable and is affine over S . If G is of finite presentation over S , then F is locally of finite presentation over S .

b) Let H be an S -group prescheme of multiplicative type, and finite over S . Then $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable. It is affine over S , and if G is of finite type (resp. of finite presentation) over S , then so is $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$.

Remark 3.13. Except for the precision that $\underline{\text{Hom}}_{S\text{-gr}}$ is affine, and in the case where G is of finite presentation over S (which will suffice for us), 3.12 is an immediate consequence of the theory of Hilbert Schemes (A. Grothendieck, *Techniques de construction et théorèmes d'existence en Géométrie Algébrique: IV Les Schémas de Hilbert*, Séminaire Bourbaki, May 1961, no. 221). It even suffices that G be quasi-projective over S ; in case a), one can also represent the larger functor

$$F'(T) = \left\{ \begin{array}{l} \text{set of group sub-preschemes of } G_T, \\ \text{flat, proper and of finite presentation over } T, \end{array} \right\}$$

(the canonical monomorphism $F \longrightarrow F'$ is an “open immersion”, as follows from criterion 3.1 and X 4.7 b), so the representability of F' implies that F is representable by an open subset of F'); in case b), one can restrict oneself to supposing that H is projective and of finite presentation over S . In both cases, one obtains a functor locally of finite presentation over S . In the present exposé, 3.12 is only a technical lemma for proving a key result in the following no. , so we shall sketch an easy direct proof of 3.12, not using Hilbert schemes.

Let us first prove b). It will suffice that we prove that $\underline{\text{Hom}}_S(H, G)$ is representable (independently of any group structure on H, G), and has the additional properties stated for $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$, for, using also the same result for $\underline{\text{Hom}}_S(H \times_S H, G)$, one describes the subfunctor $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ of $\underline{\text{Hom}}_S(H, G)$ by a finite projective limit (in fact, by means of fibered products) involving G , $\underline{\text{Hom}}_S(H, G)$ and $\underline{\text{Hom}}_S(H \times_S H, G)$, which we leave the reader to describe explicitly. On the other hand, one will have

$$H = \text{Spec}(\mathcal{B}),$$

where $\mathcal{B}$ is a sheaf of algebras on S that is locally free as a sheaf of modules (this is the only hypothesis on H that we shall have to retain). Since the representability question considered is local on S , we can suppose S affine with ring A . On the other hand, one will have $G = \text{Spec}(C)$, where C is an A -algebra. When $G = S[t] = G_0$, t an indeterminate, the functor $\underline{\text{Hom}}$ is nothing other than

$$T \longmapsto \Gamma(T, \mathcal{B}_T),$$

which is representable ($\mathcal{B}$ being locally free) by the vector bundle $\mathbf{V}(\mathcal{B}^\vee)$, where $\mathcal{B}^\vee$ is the sheaf of modules dual to $\mathcal{B}$. When $G = S[(t_i)]$, with (t_i) a family (not necessarily finite) of indeterminates, one will therefore have $G = G_0^I$ (product over S of a family of copies of G_0), which is representable by the affine scheme

$$\underline{\text{Hom}}_S(H, G)^I = \mathbf{V}(\mathcal{B}^\vee)^I = \mathbf{V}(\mathcal{B}^{\vee(I)}).$$

In the general case, G will be isomorphic to a closed subscheme of a scheme of the form $S[(t_i)]$, i.e. C will be a quotient of an A -algebra of the form $A[(t_i)]$. Let (F_j) be a system of generators of the ideal by which one takes the quotient. Suppose $\mathcal{B}$ free of rank n , which is permissible on covering S by smaller affine open subsets. Choose a basis $(e_k)_{1 \leq k \leq n}$ of $\mathcal{B}$; then, writing the n components with respect to this basis of $F_j((x_i))$, for $x_i = \sum x_{ik} e_k$ (x_{ik} indeterminate coefficients, taken in an unspecified algebra A' over A), one finds, for every F_j ,

n polynomials $F_{j,k}$ in the $(x_{i,k})_{i \in I, 1 \leq k \leq n}$, with coefficients in A . One observes at once that $\underline{\text{Hom}}_S(H, G)$ is represented by the spectrum of the quotient of the polynomial ring $A[(x_{i,k})]$ by the ideal generated by the $F_{j,k}$. This proves b) at once.

Let us prove a). For every ordinary finite commutative group M , let F_M be the subfunctor of F obtained by restricting oneself to subgroups of G_T that are of multiplicative type and of type M . One sees easily, by a gluing argument like the one used in 3.5 (which we should have stated by breaking things down a little more!), that it suffices to verify that the F_M are representable; then F will be representable by the sum prescheme of the F_M , where M runs through the set of classes of finite commutative groups up to isomorphism. (F is indeed the sum of the F_M in the category of sheaves...).

Henceforth we suppose M fixed, and shall write F instead of F_M . Let $H = D_S(M)$, consider the subfunctor $F' = \underline{\text{Imm}}_{S\text{-gr}}(H, G)$ of $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ whose value at T is the set of homomorphisms of T -groups $H_T \rightarrow G_T$ that are closed immersions. One already knows by b) that $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable by an affine S -prescheme, and using IX 6.8 one sees at once that F' is representable by an open and closed sub-prescheme of the latter, hence is also affine over S . Finally, consider the canonical morphism $F' \rightarrow F$, which associates to each monomorphism $H_T \rightarrow G_T$ the image group. One observes at once, by the definitions, that in this way F' becomes a principal homogeneous bundle (in the category of sheaves for fpqc) over F , with group $\Gamma_F = \Gamma_S \times_S F$, where $\Gamma = \text{Aut}_{\text{gr}}(M)$, whence it follows “by descent” that this morphism is representable (i.e. for every morphism $T \rightarrow F$, with T representable, $T \times_F F'$ over T is representable—in fact, representable by a principal Galois bundle under Γ). Thus, by IV 4.6.6, F is representable if and only if the quotient F'/F'' , where $F'' = F' \times_F F'$, exists in (Sch) and is universally effective for faithfully flat quasi-compact morphisms, or, what amounts to the same thing, if and only if the quotient F'/Γ exists and is universally effective for the said morphisms. Now, since F' is affine over S , we saw in V 4.1 that the condition in question is indeed satisfied. This proves the representability of F in a).

As for the addition, relating to the case where one supposes G of finite presentation over S , it follows at once from the preceding proof, taking into account that by b), F' is then locally of finite presentation over S , so that one can apply Exposé V⁽³⁾.

4. The scheme of subgroups of multiplicative type of a smooth affine group

The main result of the present exposé consists of the

Theorem 4.1. *Let S be a prescheme, G an S -group prescheme smooth and affine over S , F the functor $(\text{Sch})/S \rightarrow (\text{Ens})$ defined by*

$$F(T) = \text{set of subgroups of multiplicative type of } G_T.$$

Then the functor F is representable, and is smooth and separated over S .

Let us point out at once the following variant:

Corollary 4.2. *Let G, H be two S -group preschemes, with G smooth and affine over S , H of multiplicative type and of finite type. Then $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable, and is smooth and separated over S .*

Indeed, when H is smooth over S , so is $H \times_S G$, and one can apply 4.1 to the latter. By considering graph groups, $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ becomes a subfunctor F' of the functor F : “subgroups of multiplicative type of $H \times_S G$ ”, namely $F'(T) = \text{set of subgroups } K \text{ of multiplicative type of } (H \times_S G)_T = H_T \times_T G_T \text{ such that the homomorphism induced by the first projection}$

$$K \rightarrow H_T$$

⁽³⁾**N.D.E.** Reference to be checked/made precise.

is an isomorphism. By IX 2.9 the canonical morphism $F' \rightarrow F$ is an open immersion, so, since F is representable, so is F' , which will be representable by an open subset of F ; and since F is separated and smooth over S , so will F' be. In the case where H is finite over S , it suffices to apply 3.12 b) for the representability of $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$. When H is a product $H_1 \times_S H_2$, with H_1 smooth over S and H_2 finite over S , then $\mathrm{Hom}_{T\text{-gr}}(H_T, G_T)$ is identified with the subset of $\mathrm{Hom}_{T\text{-gr}}((H_1)_T, G_T) \times \mathrm{Hom}_{T\text{-gr}}((H_2)_T, G_T)$ formed by pairs (u_1, u_2) such that $u_1 \cdot u_2 = u_2 \cdot u_1$, whence it follows that $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$ is representable by a closed sub-prescheme of $\underline{\mathrm{Hom}}_{S\text{-gr}}(H_1, G) \times_S \underline{\mathrm{Hom}}_{S\text{-gr}}(H_2, G) = X$, as one sees by applying VIII 6.5 b), where one takes $Y = H$, $Z = G$, q_1 and q_2 being defined respectively by $(u_1, u_2) \mapsto u_1 \cdot u_2$ and $(u_1, u_2) \mapsto u_2 \cdot u_1$.

In the general case, since the question is local on S for the Zariski topology, one can suppose S affine, and H of constant type over S . Then, since H is quasi-isotrivial (X 4.5), one can find an étale surjective morphism $S' \rightarrow S$, S' affine, which splits H , i.e. such that $H' = H_{S'}$ is diagonalizable. Then the preceding result applies, for a diagonalizable group is the product of a diagonalizable torus with a finite diagonalizable group. It remains to see that the descent datum obtained on the S' -prescheme $\underline{\mathrm{Hom}}_{S'\text{-gr}}(H', G') = X'$ is effective. This is seen by the argument of X 5.4, noting that in *loc. cit.*, the hypothesis that $X' \rightarrow S'$ was separated and locally quasi-finite served only to ensure that every open subset of X' quasi-compact over S' was quasi-affine over S' . Now this property is still satisfied in the present case, as follows easily from the fact that it holds for the functor F of 4.1 (which will be seen in the course of the proof of 4.1). This (or any other variant of this little reduction) establishes the representability of $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$, and at the same time the fact that it is separated over S . It is locally of finite presentation over S , as one sees, for example (as was pointed out in 3.2), thanks to the fact that this functor “commutes with inductive limits of rings”. Finally, since this functor is formally smooth over S (by 2.1), it is smooth over S .

Remark 4.3. We have here deduced 4.2 from 4.1, which is really immediate only when H is also smooth over S . For the deduction to be made without contortions in the general case, the representability result 4.1 would have to be established without supposing G smooth over S , but only affine of finite presentation over S . (Of course, then F will in general no longer be smooth over S !). There is little doubt that 4.1 remains true under these more general hypotheses, but the proof seems to have to be more delicate (for lack of being able to invoke 3.8)^{*(4)}. Let us point out, however, that when G is a closed subgroup of an affine smooth group G' over S , then the functor F representing subgroups of multiplicative type of G is representable by a closed sub-prescheme of the prescheme representing the analogous functor F' for G' (falling under 4.1), as one sees easily by applying VIII 6.4. This also raises the question: is a group scheme G over an affine S , which is affine and of finite presentation over S , isomorphic to a group subscheme of a $\mathrm{GL}(n)_S$, for suitable n ? This is true when S is the spectrum of a field, cf. VI_B 11.11, but unfortunately false in general, even for tori, cf. 4.6. Finally, note that one could also prove 4.2 directly by exactly the same method as 4.1.

Let us now prove 4.1. Since the functor F is obviously of a local nature, one can suppose S affine, hence $S = \mathrm{Spec}(A)$, A a ring. Considering A as the inductive limit of its subrings of finite type over $\mathbf{Z}$, and noting that G comes from a smooth affine group over such a subring (EGA IV 8), one is reduced to the case where S is noetherian. For every integer $n > 0$, let T_n be the functor defined like F , but restricting oneself to subgroups H of multiplicative type of G_T such that $n \cdot \mathrm{id}_H = 0$, i.e. such that ${}_n H = H$. Order the set I of integers > 0 by the

* It is indeed proved for G flat and quasi-affine over S with connected fibers, provided one restricts oneself to central subtori of G (XV 8.8).

⁽⁴⁾**N.D.E.** In the case where H is smooth (not necessarily affine) over a locally noetherian normal base S , M. Raynaud showed that the largest representable open subset of the functor of subtori of H is a disjoint sum of smooth affine open subsets over S . This is theorem IX.9.26 in *Faisceaux amples sur les schémas en groupes et sur les espaces homogènes*, Lecture Notes Maths. 119 (1970).

divisibility relation. When m is a multiple of n , define

$$p_{n,m} : T_m \longrightarrow T_n$$

by $p_{n,m}(H) = {}_nH = \text{Ker}(n \cdot \text{id}_H)$.

In this way the T_n form a projective system of functors over S . By 3.12 a), the functors T_n are representable, affine and of finite type over S . Define likewise morphisms

$$u_n : F \longrightarrow T_n,$$

by the relation $u_n(H) = {}_nH$. In this way one obtains a morphism

$$u : F \longrightarrow T = \varprojlim_n T_n,$$

where the $\varprojlim$ is taken in the category of functors over S . But let us point out at once that, since the T_n are representable and affine over S , so is T (it will be the spectrum of the inductive limit of the quasi-coherent algebras over S defining the T_n). Of course T is not in general of finite type over S .

We shall apply 3.7 and are reduced to verifying conditions a) to d) of 3.7, which will imply that F is representable by a prescheme locally of finite type over S . It then follows from 2.1 bis that F is even smooth over S , and since F is a subfunctor of T , which is affine over S , it follows that F is separated over S (being separated over T , which is separated over S). Let us prove at once the addition invoked above, namely that every open subset U of F quasi-compact over S is quasi-affine over S : this follows from 3.11 and the fact that the T_u are affine over S .

Let us therefore verify the conditions of 3.7.

a) F is a sheaf for the faithfully flat quasi-compact topology, by descent theory SGA 1 VIII, which applies here since groups of multiplicative type over T are affine over S . It commutes with inductive limits of rings by the general groundwork EGA IV 8. Let us show that it commutes with adic projective limits of artinian local rings. When one is dealing, instead of the functor F , with the analogous functor considered in 4.2, this property is nothing other than that of IX 7.1 in the special case where A is a complete noetherian local ring, endowed with an ideal of definition for its usual topology (N.B. It is precisely here that the hypothesis G affine enters essentially). In the present case, we are reduced to proving the

Lemma 4.4. *Let A be a complete noetherian local ring, with maximal ideal $\mathfrak{m}$, G an affine group scheme over $S = \text{Spec}(A)$. For every integer $n \geq 0$, let $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, $G_n = G \times_S S_n$. For every n , let H_n be a subgroup of multiplicative type and of finite type of G_n , such that for $m \geq n$, H_n is deduced from H_m by reduction. Under these conditions, there exists a unique subgroup H of multiplicative type in G that reduces to the H_n .*

By X 3.2 there exists a group H of multiplicative type over S , necessarily of finite type and isotrivial, determined up to unique isomorphism, endowed with an isomorphism $H \times_S S_0 \simeq H_0$ (using the fact that H_0 is isotrivial, being of finite type over a field, cf. X 1.4). By X 2.1, for every n , the isomorphism $H \times_S S_0 \simeq H_0$ lifts to a unique isomorphism $H \times_S S_n \simeq H_n$. This being said, by IX 7.1 already cited, the homomorphisms $H_n \longrightarrow G_n$ come from a unique homomorphism of S -groups $u : H \longrightarrow G$.

By IX 6.6 the latter is a monomorphism since $u_0 : H_0 \longrightarrow G_0$ is. This completes the proof of 4.4.

b) The morphism $u : F \longrightarrow T$ is a monomorphism. This follows from the density theorem IX 4.7, in the form of corollary 4.8 b). One should take care that it is essential, for the application we make of it here, to have this result over a base that is not necessarily noetherian. (N.B. Since the functor T does not commute with inductive limits of rings, it is not possible to reduce to that case a priori).

b') The morphism $u : F \longrightarrow T$ is infinitesimally étale, in other words one has the

Lemma 4.5. *Let A be an artinian local ring with residue field k , $S = \operatorname{Spec}(A)$, I an ideal $\subset \operatorname{rad}(A)$, $S' = \operatorname{Spec}(A/I)$, G an S -group prescheme smooth over S , $G' = G \times_S S'$; give ourselves a subgroup H' of multiplicative type of G' and, for every integer $n > 0$, a subgroup $H(n)$ of multiplicative type of G , in such a way that*

- 1°) *for every multiple m of n , $H(n) = {}_n H(m)$, and*
- 2°) *$H(n)' = {}_n H'$.*

Under these conditions, there exists one and only one subgroup H of multiplicative type of G such that ${}_n H = H(n)$ for every n .

Uniqueness is already contained in b). For existence, an immediate induction reduces us to the case where $J\mathfrak{m} = 0$, $\mathfrak{m}$ being the maximal ideal of A . Let $k = A/\mathfrak{m}$, $S_0 = \operatorname{Spec}(k)$, $G_0 = G \times_S S_0$, $\mathfrak{g}_0$ the Lie algebra of G_0 , $\mathfrak{h}_0$ that of H_0 . One has a canonical group isomorphism:

$$\mathfrak{g}_0 \otimes_k J \simeq \operatorname{Ker}(G(S) \longrightarrow G(S_0)),$$

cf. Exp. III. By IX 3.6 bis and 3.7, there exists a subgroup H of multiplicative type of G reducing to H' , and such an H is determined modulo inner automorphism by an element of $N = \operatorname{Ker}(G(S) \longrightarrow G(S_0))$. Thus the set P of liftings H of H_0 is a principal homogeneous set under the group N/M , where M is the group of $g \in N$ such that $\operatorname{int}(g) \cdot H = H$.

Now one sees easily (Exp. III) that this subgroup is nothing other than the vector subspace of $\mathfrak{g}_0 \otimes_k J$ formed by the invariants under H_0 , when H_0 acts on $\mathfrak{g}_0 \otimes_k J$ by the representation induced by the adjoint representation of G_0 . Likewise, the set $P(n)$ of liftings of $H(n)_0 = {}_n H_0$ is a principal homogeneous set under $N/M(n)$, where $M(n)$ is the vector subspace of $\mathfrak{g}_0 \otimes_k J = N$ formed by the invariants under ${}_n H_0$. Using the density theorem Exp. IX 4.7, one sees easily that for large n (in the sense of the order relation put on the set of integers $n > 0$, namely the divisibility relation) one has $M = M(n)$. Consequently the natural map $H \mapsto {}_n H$ from P to $P(n)$, which is compatible with the actions of N and hence with the homomorphism $N/M \longrightarrow N/M(n)$ on the groups of operators, is bijective for large n . The conclusion of 4.5 follows at once.

To verify c) and d) of 3.7, we use 3.8, which reduces us to the verification of c') and d') below.

c') The T_n are smooth over S , and the transition morphisms $p_{n,m} : T_m \longrightarrow T_n$ are smooth.

This is nothing other than 2.2 bis.

d') With the notation of 3.8, a point ξ of F with values in a field k over S is nothing other than a subgroup H_0 of multiplicative type of $G_0 = G_k$. Taking up again the argument of b') above, one sees that the integer $d(\xi)$ considered in 3.8 is nothing other than the dimension of $\mathfrak{g}_0/\mathfrak{g}_0^{H_0}$, where $\mathfrak{g}_0$ is the Lie algebra of G_0 and $\mathfrak{g}_0^{H_0}$ is the vector subspace of invariants under H_0 . It is therefore a finite integer, i.e. condition d') 1°) of 3.8 holds. With the notation of d') 2°, giving a morphism $v : X \longrightarrow F$ amounts to giving a subgroup H of multiplicative type of G_X . For $x \in X$, the integer $d(\xi_x)$ is then nothing other than the dimension of $(\mathfrak{g} \otimes \kappa(x))^{H_x}$, where $\mathfrak{g}$ is the sheaf of Lie algebras of G_X (which is locally free of finite type on X , for G_X is smooth over X , and $\mathfrak{g} \otimes \kappa(x)$ is nothing other than the Lie algebra of the fiber G_x of G_X at x), and H_x is the fiber of H at x . Now, since $\mathfrak{g}$ is a locally free module over S on which the group H of multiplicative type acts, one sees at once that the subfunctor $\mathfrak{g}^H$ of invariants under H is given by a subsheaf locally a direct summand, hence locally free, of $\mathfrak{g}$. (By descent, one is reduced to the case where H is diagonalizable, and where one applies Exp. I 4.7.3, noting that the subsheaf of invariants corresponds to the component of degree zero). Consequently $d(\xi_x) = \operatorname{rank}$ at x of $\mathfrak{g}^H$, hence it is a locally constant function of x . This completes the proof of condition d').

We have thus verified the conditions of 3.7, which completes the proof of 4.1.

Remark 4.6. When $G = \operatorname{GL}(n)_S$, one can give a direct proof of 4.1 that is distinctly simpler and more explicit, using I 4.7.3. The proof shows in addition that, in this case, the moduli scheme is a sum scheme of a family of affine schemes over S . Proceeding as was said in 4.3,

one deduces the same result whenever G is a closed subgroup of a group of the form $\mathrm{GL}(n)_S$. One should beware, however, of believing that the preschemes representing the functors in 4.1 or 4.2 are always sums of a family of affine schemes over S . Let, for example, H be a group of multiplicative type and of finite type over a locally noetherian prescheme S ; then by X 5.11, H is isotrivial if and only if $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, \mathbf{G}_m)$ is a sum of a family of affine preschemes over S , but it was pointed out (X 1.6) that non-isotrivial tori H (of relative dimension 2) can exist; for such a torus, $\underline{\mathrm{Hom}}_{S\text{-gr}}(H, \mathbf{G}_m)$ is therefore not a sum of affine S -preschemes over S , and one sees likewise that the “dual” twisted constant group $\underline{\mathrm{Hom}}_{S\text{-gr}}(\mathbf{G}_m, H)$ is not such a sum either (for if two commutative twisted constant groups of finite presentation R, R' are dual to each other, $R' = \underline{\mathrm{Hom}}_{S\text{-gr}}(R, \mathbf{Z}_S)$, one sees easily that one is isotrivial if and only if the other is). This last point also shows that such an H is not isomorphic to a subgroup of a group of the form $\mathrm{GL}(n)_S$; more precisely, one can show that a group of multiplicative type and of finite type over a connected locally noetherian S is isomorphic to a subgroup of a group of the form $\mathrm{Aut}_{(\mathrm{Mod})}(\mathcal{E})$ (with $\mathcal{E}$ a locally free module of finite type over S) if and only if it is isotrivial. Finally, taking $G = H \times \mathbf{G}_m$ in the two preceding examples, one finds an example where the prescheme representing the functor F of 4.1 is not a sum of a family of affine S -preschemes over S (with G a torus of relative dimension 3, if desired).

5. First corollaries of the representability theorem

Let H, G be as in 4.2. Put

$$M = \underline{\mathrm{Hom}}_{S\text{-gr}}(H, G),$$

which is a smooth S -prescheme, separated over S , by 4.2. Note that G acts on the functor $M = \underline{\mathrm{Hom}}_{S\text{-gr}}(H, G)$ by

$$(g, u) \longmapsto \mathrm{int}(g) \circ u,$$

whence a canonical morphism

$$(x) \quad G \times_S M \longrightarrow M \times_S M,$$

whose components are the preceding morphism $G \times_S M \longrightarrow M$, and the second projection $G \times_S M \longrightarrow M$.

Corollary 5.1. *The preceding morphism (x) is smooth.*

This follows from 4.2 and 2.3. This statement is equivalent to the following:

Corollary 5.2. *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S -groups. Then the subfunctor $\underline{\mathrm{Transp}}(u_1, u_2)$ of G (cf. 2.4) is representable by a closed sub-prescheme of G , smooth over S .*

It remains only to prove that $\underline{\mathrm{Transp}}(u_1, u_2) \longrightarrow G$ is indeed a closed immersion, which follows from the fact that it is the kernel of a pair of morphisms $G \rightrightarrows M$ (namely $g \longmapsto \mathrm{int}(g)u_1$ and the “constant” morphism $g \longmapsto u_2$), and from the fact that M is separated over S . In particular:

Corollary 5.3. *Let $u : H \longrightarrow G$ be a morphism of S -groups. Then $\underline{\mathrm{Centr}}_G(u) = \underline{\mathrm{Transp}}(u, u)$ is representable by a closed group sub-prescheme of G , smooth over S . In addition,*

$$G/\underline{\mathrm{Centr}}_G(u)$$

is representable by an open sub-prescheme of M .

It remains to prove this last point. Now the morphism

$$g \longmapsto \mathrm{int}(g) \circ u$$

from G to M is smooth of finite type by 5.2, hence is an open morphism (EGA IV 6.6), and if U denotes its image, with the structure induced by M , the induced morphism $G \longrightarrow U$ is smooth, surjective, of finite type, hence covering for the faithfully flat quasi-compact topology. Moreover, it is obvious that the preceding morphism $G \longrightarrow M$ makes G a formally principal

homogeneous sheaf under $\underline{\text{Centr}}_G(u)_M$, which implies that the sheaf $G/\underline{\text{Centr}}_G(u)$ is indeed representable by U .

Corollary 5.4. *Let $u_1, u_2 : H \rightrightarrows G$ be two homomorphisms of S -groups. The following conditions are equivalent:*

- (i) u_1 and u_2 are locally conjugate for the étale topology.
- (i bis) u_1, u_2 are locally conjugate in the sense of the faithfully flat quasi-compact topology.
- (ii) For every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, the morphisms $u_{1,\bar{s}}, u_{2,\bar{s}} : H_{\bar{s}} \rightrightarrows G_{\bar{s}}$ are conjugate by an element of $G(\overline{\kappa(s)})$.
- (ii bis) The structural morphism $\underline{\text{Transp}}(u_1, u_2) \rightarrow S$ is surjective.
- (iii) $\underline{\text{Transp}}(u_1, u_2)$ is a torsor under the action of the smooth S -group prescheme of finite type $\underline{\text{Centr}}_G(u_1)$.

(i) $\Rightarrow$ (i bis) and (ii) $\Rightarrow$ (ii bis) are trivial (the second thanks to the Nullstellensatz); moreover (i bis) $\Rightarrow$ (ii) by the “finite extension principle” (EGA IV 9). On the other hand, (ii bis) $\Rightarrow$ (iii) thanks to the fact that $\underline{\text{Transp}}(u_1, u_2)$ is smooth over S , hence flat over S , and of finite type, hence quasi-compact over S , so it is faithfully flat quasi-compact over S if and only if its structural morphism is surjective. Since, on the other hand, it is formally principal homogeneous under $\underline{\text{Centr}}_G(u_1)$, which is faithfully flat quasi-compact over S , one sees that this last condition is also equivalent to saying that $\underline{\text{Transp}}(u_1, u_2)$ is a torsor under $\underline{\text{Centr}}_G(u_1)$ (understood: in the sense of the faithfully flat quasi-compact topology). Finally (iii) $\Rightarrow$ (i) thanks to the fact that $\underline{\text{Transp}}(u_1, u_2)$ is smooth over S and to “Hensel’s lemma” in the form 1.10.

Remark 5.5. For fixed $u_1 = u : H \rightarrow G$, the functor $(\text{Sch})/S \rightarrow (\text{Ens})$ that associates to every T over S the set of homomorphisms of T -groups $u_2 : H_T \rightarrow G_T$ that are conjugate to $u_T : H_T \rightarrow G_T$ locally for the étale topology is precisely representable by the open subset of M , isomorphic to $G/\underline{\text{Centr}}_G(u)$, considered in 5.3.

Let us sketch the variants of the preceding results obtained by applying 4.1 instead of 4.2. Thus let G be a group prescheme smooth and affine over S , and now denote by M the smooth S -prescheme, separated over S , representing the functor considered in 4.1. One again has actions of G on M :

$$(g, H) \mapsto \text{int}(g)(H),$$

whence, as above, a morphism

$$(x \text{ bis}) \quad G \times_S M \rightarrow M \times_S M.$$

Corollary 5.1 bis. *The preceding morphism is smooth.*

Follows from 4.1 and 2.3 bis. One again concludes:

Corollary 5.2 bis. *Let H_1, H_2 be two subgroups of multiplicative type of G . Then the subfunctor $\underline{\text{Transp}}_G(H_1, H_2)$ of G is representable by a closed sub-prescheme of G , smooth over S .*

In particular:

Corollary 5.3 bis. *Let H be a subgroup of multiplicative type of G . Then the group subfunctor $\underline{\text{Norm}}_G(H)$ of G is representable by a closed sub-prescheme of G , smooth over S . In addition, the quotient $G/\underline{\text{Norm}}_G(H)$ is representable by an open sub-prescheme of M .*

Corollary 5.4 bis. *Let H_1, H_2 be two subgroups of multiplicative type of G . The following conditions are equivalent:*

- (i) H_1 and H_2 are locally conjugate for the étale topology.
- (i bis) H_1 and H_2 are locally conjugate for the faithfully flat quasi-compact topology.

- (ii) For every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, the subgroups $H_{1,\bar{s}}, H_{2,\bar{s}}$ of $G_{\bar{s}}$ are conjugate by an element of $G(\overline{\kappa(s)})$.
- (ii bis) The structural morphism $\text{Transp}(H_1, H_2) \rightarrow S$ is surjective.
- (iii) $\text{Transp}(H_1, H_2)$ is a principal homogeneous bundle under the action of the smooth S -group prescheme of finite type $\text{Norm}_G(H_1)$.

Remark 5.5 bis. Remark 5.5 also carries over to the present case.

Remark 5.6. Note that to establish result 5.2, and consequently also the first assertion in 5.3, as well as 5.4, the reference to 4.2 can be replaced by a reference to VIII 6.4, whose proof is much easier. This also shows that the hypothesis G affine over S is unnecessary there. In addition, in no. 6 we shall show how a variant of this method allows these results to be extended to the case of certain groups H more general than groups of multiplicative type. These same observations extend to variants 5.2 bis etc. On the other hand, result 5.8 below uses essentially all the hypotheses made (in particular G affine and smooth over S , H of multiplicative type), and the lecturer knows no other proof of it than via the representability theorems 4.1 or 4.2*.

5.7. — Since the morphism (x), resp. (x bis), is smooth, hence open, its image is open. Let R be this image, endowed with the structure induced by $M \times_S M$; one observes easily that R is an *equivalence relation* in M , which is moreover nothing other than that described explicitly in 5.4, resp. 5.4 bis. It would be interesting to know whether the quotient sheaf M/R (which is formally étale over S) is representable (it is then representable by an étale prescheme over S); it is so, for example, if S is the spectrum of a field. One will note, moreover, that it can happen that R is not closed in $M \times_S M$ (even if G is quasi-finite over $S \dots$), which means (when M/R is representable) that M/R is then not separated over S .

Theorem 5.8. Let S be a prescheme, G an S -prescheme smooth and affine over S , $s \in S$. Then:

- a) For every subgroup H_0 of multiplicative type of G_s , there exists an étale morphism $S' \rightarrow S$, a point s' of S' above s such that the residue extension $\kappa(s')/\kappa(s)$ is trivial, and a subgroup H' of multiplicative type of $G' = G \times_S S'$ such that $H'_{s'} = H_0 \otimes_{\kappa(s)} \kappa(s')$.
- b) For every group homomorphism $u_0 : H_s \rightarrow G_s$, where H is an S -group prescheme of multiplicative type and of finite type, there exists an étale morphism $S' \rightarrow S$, a point s' of S' above s such that the residue extension $\kappa(s')/\kappa(s)$ is trivial, and a group homomorphism $u' : H' \rightarrow G'$ such that $u'_{s'} : H'_{s'} \rightarrow G'_{s'}$ is equal to $u_0 \otimes_{\kappa(s)} \kappa(s')$.

This follows from 4.1, resp. 4.2, and Hensel's lemma in the form 1.10.

Proposition 5.9. Let S be a prescheme, G an S -prescheme smooth and affine over S , H a subgroup of multiplicative type of G . Consider $\text{Centr}_G(H) \hookrightarrow \text{Norm}_G(H)$, which by 5.3 and 5.3 bis are closed group sub-preschemes of G , smooth over S . Then the first group is an open and closed sub-prescheme of the second, and the quotient sheaf

$$W_G(H) = \text{Norm}_G(H) / \text{Centr}_G(H)$$

is representable by an open group sub-prescheme of $\text{Aut}_{S\text{-gr}}(H)$; hence it is an S -group prescheme quasi-finite, étale and separated over S .

Indeed, consider the obvious homomorphism

$$\theta : \text{Norm}_G(H) \rightarrow \text{Aut}_{S\text{-gr}}(H),$$

whose kernel is by definition $\text{Centr}_G(H)$. Since $\text{Aut}_{S\text{-gr}}(H)$ is representable by an S -group prescheme étale and separated over S (X 5.10), its identity section is an open and closed

*The situation has changed since the writing of this text, cf. XV and XIX no. 6.

immersion, so its inverse image under θ is an open and closed subgroup of $\underline{\text{Norm}}_G(H)$. I say, in addition, that θ is a *smooth morphism*: this follows indeed formally from the definitions, and from the fact that $\underline{\text{Norm}}_G(H)$ is smooth over S and $\underline{\text{Aut}}_{S\text{-gr}}(H)$ is étale over S . One concludes as in 5.3 that the image of θ is an open subset U of $\underline{\text{Aut}}_{S\text{-gr}}(H)$ and that, endowed with the induced structure, U represents the quotient sheaf $\underline{\text{Norm}}_G(H)/\underline{\text{Centr}}_G(H)$. The latter is therefore étale and separated over S , since $\underline{\text{Aut}}_{S\text{-gr}}(H)$ is, and it is quasi-finite over S , being quasi-compact over S as the image of $\underline{\text{Norm}}_G(H)$, which is so. This completes the proof of 5.9.

Corollary 5.10. *For every $s \in S$, let*

$$w(s) = \text{rank } \underline{\text{Norm}}_{G_s}(H_s)/\underline{\text{Centr}}_{G_s}(H_s)$$

(which is also the index of $\underline{\text{Centr}}_{G(k)} H(k)$ in $\underline{\text{Norm}}_{G(k)} H(k)$, where k is an algebraic closure of $\kappa(s)$). Then the function $s \mapsto w(s)$ is lower semicontinuous. For it to be constant near the point s , it is necessary and sufficient that $W_G(H)$ be finite over S near s .

Indeed, for every S -prescheme W that is étale, of finite type and separated over S , it is true that the function $s \mapsto w(s) = \text{rank } [W_s : \kappa(s)]$ is lower semicontinuous, and that it is constant near the point s if and only if W is finite over S near s (a fact pointed out in SGA 1, I 10.9, whose proof, which offers no difficulty, will be found in EGA IV^{*}).

Remark 5.11. ⁽⁵⁾Let G be a group prescheme affine and smooth over S , H a subgroup of multiplicative type; then 5.3 and 5.3 bis imply that the quotients

$$G/\underline{\text{Centr}}_G(H) \quad \text{and} \quad G/\underline{\text{Norm}}_G(H)$$

are smooth preschemes over S and *quasi-affine* over S , since in either case the moduli prescheme M into which the quotient is embedded is such that every open subset U of M quasi-compact over S is quasi-affine over S (3.11). We shall moreover see in the following exposé⁽⁶⁾ that for every U as above, the schematic closure $\overline{U}$ of U in M is even affine over S , which shows that the quotients considered are affine over S provided the open subset U in the corresponding moduli scheme (of group homomorphisms from H to G , resp. of subgroups of multiplicative type of G) is closed in M . This is, for example, the case if H is a “maximal torus” in the second case considered, or when S is the spectrum of a field, cf. XII.5.4⁽⁷⁾. I do not know whether the quotients considered are affine over S in general.

6. On a rigidity property for homomorphisms of certain group schemes, and the representability of certain transporters^{*}

Proposition 6.1. *Let S be a locally noetherian prescheme, H a commutative S -group prescheme of finite type over S , E a set of integers > 0 stable under multiplication, and suppose the following conditions hold:*

- a) *For every $n \in E$, the subgroup ${}_n H = \text{Ker}(n \cdot \text{id}_H)$ is finite and flat over S .*
- b) *For every $s \in S$, the family of ${}_n H_s$ ($n \in E$) is schematically dense in H .*

^{*}EGA IV 15.5.1 and 18.10.7.

⁽⁵⁾**N.D.E.** For a generalization to the non-affine case, see M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes Math. 119 (1970), IX.2.8.

⁽⁶⁾**N.D.E.** Cf. remark XII.5.7. Let us mention here the following generalization. Without any affineness hypothesis on G , if H is a torus and S is locally noetherian, U is affine and smooth by a theorem of Raynaud (*loc. cit.*, IX.2.6 and IX.2.8). Indeed, $G/\underline{\text{Norm}}_G(H)$ is an open subset of M and the largest representable open subset of M is a disjoint sum of smooth affine open subsets over S .

⁽⁷⁾**N.D.E.** Under the hypothesis that G has locally constant reductive rank.

^{*}The present no. does not use the results of Nos. 3, 4, 5; its natural place would be in VI_B.

Let Y be a closed sub-prescheme of H . Then the subfunctor $\prod_{H/S} Y/H$ of S (compare VIII 6) is representable by a closed sub-prescheme T of S , and if S is noetherian, there exists an $n \in E$ such that

$$(x) \quad \prod_{H/S} Y/H = \prod_{nH/S} Y \cap_n H/S.$$

Indeed, by VIII 6.4, since by condition a) $_nH$ is finite and flat and a fortiori “essentially free” over S , it follows that the second member of (x) is representable by a closed sub-prescheme T_n of T . Of course, for $n \in E$, E ordered by divisibility, the T_n form a decreasing family of closed sub-preschemes of S , so if S is noetherian (which we can suppose), it is stationary for large n .

Let T be the value of T_n for large n ; I say that T does indeed represent the first member of (x), which will complete the proof. One is reduced to proving that if S' is a prescheme over S such that $(Y \cap_n H)_{S'} = (_nH)_{S'}$ for every $n \in E$, i.e. such that $Y_{S'} \supset _nH_{S'}$ for every $n \in E$, then $Y_{S'} = H_{S'}$. Now this is indeed the case, for by IX 4.4 the family of $_nH_{S'}$ is schematically dense in $H_{S'}$, taking account of conditions a) and b).

Theorem 6.2. *Let S be a prescheme, H, E as in 6.1 satisfying conditions a), b), $u : H \rightarrow G$ a homomorphism of S -groups from H to an S -group prescheme locally of finite type over S , and finally K a closed group sub-prescheme of H . Consider the subfunctor $\text{Transp}_G(u, K)$ of G (cf. 2.5). Then the latter is representable by a closed sub-prescheme of G , and if G is noetherian (for example S noetherian and G of finite type over S), there exists an integer $n \in E$ such that $\text{Transp}_G(u, K) = \text{Transp}_G(u|_{_nH}, K)$. If finally G is smooth over S , and K smooth over S or of multiplicative type, and if H, E satisfy the following condition, stronger than a):*

a') For every $n \in E$, the subgroup $_nH$ of H is of multiplicative type, then $\text{Transp}_G(u, K)$ is smooth over S .

Indeed, consider the G -group $H_G = H \times_S G$, which obviously satisfies the conditions of 6.1 (S replaced by G , and H by H_G), and the closed sub-prescheme Y of H_G , inverse image of $K \subset G$ under

$$H_G = H \times_S G \rightarrow G$$

defined by $(h, g) \mapsto \text{int}(g) \cdot u(h)$. Then $\text{Transp}_G(u, K)$ is nothing other than $\prod_{H_G/G} Y/H_G$ (compare VIII, examples 6.5 e)). Thus the first assertions follow from 6.1, and moreover one sees that for every quasi-compact open subset U of G , there exists $n \in E$ such that $\text{Transp}_G(u, K)$ and $\text{Transp}_G(u|_{_nH}, K)$ have the same trace on U . To verify the last assertion of 6.2, one can therefore replace H by an $_nH$, and then it suffices to apply 2.5, which applies since $_nH$ is supposed of multiplicative type over S .

Remark 6.3. The preceding proof uses only the very elementary result VIII 6.4, and in addition (for the last part) 2.5, that is, when K is smooth over S , the infinitesimal result IX 3.6, hence the vanishing of the cohomology of groups of multiplicative type. One will note that in the most important cases (cf. 6.7) one can even suppose the $n \in E$ prime to the residue characteristics of S , i.e. invertible in $\mathcal{O}_S$, hence the $_nH$ finite étale over S , and then the cohomological result invoked is practically trivial, so 6.2 is then independent of the theory of groups of multiplicative type.

6.4. — One sees as usual that 6.1 extends to the case where one has an S -prescheme S' over S (not necessarily locally noetherian), and a closed sub-prescheme Y' of $H' = H_{S'}$, provided $Y' \rightarrow H'$ is of finite presentation, i.e. the ideal defining Y' of finite type: then $\prod_{H'/S'} Y'/H'$ is representable by a closed sub-prescheme T' of S' , such that $T' \rightarrow S'$ is of finite presentation, and if S' is quasi-compact, there exists an $n \in E$ such that the relation analogous to (x) holds. It also follows that the first statement in 6.2 holds without supposing G locally of finite type over S , provided the immersion $K \rightarrow G$ is of finite presentation.

6.5. — As announced in 5.6, theorem 6.2 allows one to extend to group preschemes satisfying a') and b) above certain results established by another method and under more restrictive conditions for groups of multiplicative type. This is so for results 5.2, the beginning of 5.3, 5.4 and the bis variants of the preceding results 5.9 and 5.10. It is likewise so for the results of IX, Nos. 5 and 6, excluding IX 6.8 (already false for a homomorphism of abelian schemes $u : H \rightarrow G$ over the spectrum S of a discrete valuation ring with residue characteristic $p > 0$: it can indeed happen that $\text{Ker } u$ has as generic fiber the identity group, and as special fiber a radicial group not reduced to the identity group).

6.6. — We have just given an example of a rigidity property for groups of multiplicative type that is not shared by abelian schemes. Another example, extremely important, is the fact that the existence theorem for infinitesimal extensions of homomorphisms IX 3.6 no longer holds when H is an abelian scheme. Thus, an abelian scheme $\neq 0$ over a field admits nontrivial infinitesimal variations, unlike what happens for a group of multiplicative type—which is the infinitesimal aspect of the fact that there exists a “moduli theory” (moreover far from complete) for abelian varieties, whereas moduli theory for groups of multiplicative type is empty. Another “global” aspect of this infinitesimal difference is that if H is an abelian scheme over a locally noetherian S , and G a commutative S -group prescheme locally of finite type over S , then one can show that $\underline{\text{Hom}}_{S\text{-gr}}(H, G)$ is representable by a prescheme locally of finite type over S , but, unlike what happens for H of multiplicative type, this prescheme is *not étale* over S , but only unramified over S . Thus, if S is, for example, the spectrum of a complete discrete valuation ring, H and G abelian schemes over S , there can exist homomorphisms $H_0 \rightarrow G_0$ on the special fibers that do not come “by specialization” from a homomorphism on the generic fibers.

6.7. — Theorem 6.2 applies whenever H is an abelian scheme over S , or more generally an extension of such a scheme by a torus. Indeed, since the question is local on S , one can suppose that there exists a prime number ℓ prime to the residue characteristics of S , and one sees that it then suffices to take for E the set of powers of ℓ to satisfy conditions a') and b).

An argument analogous to that of 6.1 gives us the

Theorem 6.8. *Let X be a prescheme smooth over S , with nonempty connected geometric fibers. Then for every closed sub-prescheme Y of X , the functor $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme T of S . If Y is of finite presentation over X , then T is of finite presentation over S .*

Since $f : X \rightarrow S$ is faithfully flat locally of finite presentation, it is covering for the fpqc topology. Since, on the other hand, $T = \prod_{X/S} Y/X$ is obviously a subsheaf of S (for the fpqc topology), it follows that the question of representability of T by a closed sub-prescheme of S is of a local nature on S for the fpqc topology, and the same is true of the question of deciding whether T is of finite presentation over S . On making then the base change $S' \rightarrow S$, with $S' = X$, one is reduced to the case where X admits a section e over S . One can in addition suppose S affine and a fortiori quasi-compact. One then has:

Corollary 6.9. *Under the conditions of 6.8, suppose that S is quasi-compact and that X admits a section e over S . For every integer $n \geq 0$, let X_n be the sub-prescheme of X that is the infinitesimal neighborhood of order n of the section e . Suppose Y of finite presentation over X . Then there exists an integer $n \geq 0$ such that one has $\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n$ (where $Y_n = Y \cap X_n$).*

This corollary does indeed imply 6.8 when Y is of finite presentation over X , thanks to VIII 6.4, for, since X is smooth over S , X_n is finite and locally free over S and a fortiori is “essentially free” over S in the sense of VIII 6.1, hence $\prod_{X_n/S} Y_n/X_n$ is representable by a closed sub-prescheme of S . In addition, the proof in *loc. cit.*, or the reduction to the

noetherian case, shows us immediately that the said closed sub-prescheme of S is of finite presentation over S .

Let us first prove 6.9, hence 6.8, when S is noetherian. Let $T_n = \prod_{X_n/S} Y_n/X_n$; then the T_n form a decreasing sequence of closed sub-preschemes of S , and since S is noetherian, this sequence is stationary. Let $R = \bigcap_{n \geq 0} \prod_{X_n/S} Y_n/X_n$ be their common value for large n ; one obviously has $T \subset R$, and it suffices to establish that one has $R \subset T$. On making the base change $R \rightarrow S$, one is reduced to the case where $R = S$, i.e. $Y_n = X_n$ for every n , i.e. $Y \supset X_n$ for every n , and to proving then $T = S$, i.e. $Y = X$. Now $Y \supset X_n$ for every n implies (thanks to the fact that X is locally noetherian) that Y is, near every point of $e(S)$, an induced open sub-prescheme of X , so there exists an induced open subset U of X , containing $e(S)$, such that $U \subset Y$. By IX 4.3, since the fibers of X/S are integral, U is schematically dense in X , hence (Y being a closed sub-prescheme containing U) one has $Y = X$. This proves 6.9, hence 6.8, in this case.

The general case proceeds by reduction to the preceding case. For every $s \in S$, there exists an affine open neighborhood U of s and an affine open neighborhood V of $e(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U , and if S' is an affine open neighborhood of s contained in $e^{-1}(V) \cap f(V)$, and $X' = V \cap f^{-1}(S')$, then X' and S' are affine open subsets of X , resp. S , and X'/S' admits a section. Because of the local nature of 6.8 and 6.9, one can suppose $S = S'$. I say that one then has $\prod_{X/S} Y/X = \prod_{X'/S} Y'/X'$, where $Y' = Y \cap X'$; indeed, by IX 4.6, X' is schematically dense in X (at least when X is quasi-separated over S , hence X' is retrocompact in X ; but in fact one can show without difficulty that IX 4.6 remains valid without the retrocompactness hypothesis), and likewise for every base change $S_1 \rightarrow S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$, whence at once the asserted equality. This reduces us to the case where $X = X'$, so one can suppose S and X affine.

In addition, if $X = \text{Spec}(B)$ and if J is the ideal of B defining Y , J is the inductive limit of its subideals of finite type, so Y is an intersection of closed sub-preschemes Y_i of X that are of finite presentation over S , and consequently $\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X$, which reduces us, to prove 6.8, to the case where Y is of finite presentation over X . It then suffices to prove 6.9 with S and X affine. But then X and Y over S come by the base change $S \rightarrow S_0$ from an analogous situation X_0 and Y_0 over S_0 , with S_0 noetherian, which reduces us to the case where S is noetherian, which has already been treated. This completes the proof of 6.8 and 6.9.

Corollary 6.10. *Let X be a smooth S -group prescheme of finite presentation, with connected fibers, Y a group prescheme of finite presentation over S , $i : Y \rightarrow X$ a monomorphism of S -group preschemes, thus making Y a subgroup of X . Then $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme of finite presentation of S . If S is quasi-compact, denoting for every integer $n \geq 0$ by X_n the infinitesimal neighborhood of order n of the identity section of X , and putting $Y_n = X_n \cap Y$, one has for large enough n : $\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n$.*

The proof is essentially that of 6.9. Note that the identity sections of X and Y induce bijective immersions $S \rightarrow X_n$ and $S \rightarrow Y_n$, hence induce isomorphisms of S_{red} with $(X_n)_{\text{red}}$ and $(Y_n)_{\text{red}}$, which implies that the injection $Y_n \rightarrow X_n$ is proper, hence, being a monomorphism of finite presentation, is a closed immersion. Consequently, by VIII 6.4 already used, $\prod_{X_n/S} Y_n/X_n$ is representable by a closed sub-prescheme of S of finite presentation over S , and it therefore remains to prove the last assertion of 6.10 in the case where one supposes in addition S affine. One is again immediately reduced to the case where S is noetherian, and one is reduced to proving that one then has $R = T$ (with the notation of the proof of 6.9), or again that $Y \supset X_n$ for every n implies $Y = X$. Now the hypothesis implies that $i : Y \rightarrow X$ is étale at the points of the identity section of Y over S , so Y is smooth over S at the points of the identity section, whence it follows that the open subset Y' of points of Y at which Y is smooth over S is an induced open subgroup of Y . Then $Y' \rightarrow X$ is

an étale monomorphism by X 3.5, hence an open immersion; but since the fibers of X are connected and every open subgroup of an algebraic group is also closed, it follows that this is a surjective open immersion, i.e. an isomorphism. Thus $Y' = X$ and a fortiori $Y = X$, which completes the proof of 6.10.

Proceeding as in VIII 6.5, one concludes from 6.10:

Corollary 6.11. *Let G be an S -group prescheme locally of finite type and quasi-separated over S , H a group prescheme smooth of finite presentation over S with connected fibers, $i : H \rightarrow G$ a monomorphism of S -groups, thus making H a subgroup of G . Then:*

- a) $\underline{\text{Centr}}_G(H)$ and $\underline{\text{Norm}}_G(H)$ are representable by closed sub-preschemes of G of finite presentation over G , and likewise, for every monomorphism $j : K \rightarrow G$ of finite presentation of S -group preschemes, $\underline{\text{Transp}}_G(H, K)$ is representable by a closed sub-prescheme of G of finite presentation over G .
- b) If G is quasi-compact, in the various cases considered in a), there exists an integer $n \geq 0$ such that (if H_n denotes the infinitesimal neighborhood of order n of the identity section of H) one has

$$\begin{aligned}\underline{\text{Centr}}_G(H) &= \underline{\text{Centr}}_G(H_n), \\ \underline{\text{Norm}}_G(H) &= \underline{\text{Norm}}_G(H_n), \\ \underline{\text{Transp}}_G(H, K) &= \underline{\text{Transp}}_G(H_n, K) = \underline{\text{Transp}}_G(H_n, K_n).\end{aligned}$$

One applies 6.10 to the group prescheme $X = H_G = H \times_S G$ over the base prescheme G , and to the group sub-prescheme Y that is the inverse image of the diagonal subgroup of $(G \times_S G)_G$ over G under a suitable homomorphism of G -groups from X to $(G \times_S G)_G$ (in the case of $\underline{\text{Centr}}$), resp. the inverse image of K_G under a suitable homomorphism of G -groups from X to G_G (in the case of $\underline{\text{Transp}}$). The case of $\underline{\text{Norm}}$ reduces to the transporter by taking $K = H$, the hypothesis on G ensuring that $H \rightarrow G$ is of finite presentation (hence $Y \rightarrow X$ is of finite presentation); in the case of $\underline{\text{Centr}}$, the hypothesis made on G ensures that the diagonal group of $G \times_S G$ is of finite presentation over $G \times_S G$, whence again the fact that Y is of finite presentation over X .

Remark 6.12. One can prove (using rather delicate results of EGA VI^{*}) that if X is a flat prescheme of finite presentation over S , which is proper over S or has nonempty connected geometric fibers, then for every closed sub-prescheme Y of X of finite presentation over S , $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme of S of finite presentation over S . Likewise, if X is a flat S -group prescheme of finite presentation with connected fibers, and $i : Y \rightarrow X$ a monomorphism of S -groups, with Y an S -group prescheme of finite presentation, then $\prod_{X/S} Y/X$ is representable by a closed sub-prescheme of S of finite presentation over S . In particular, 6.11 a) remains valid on replacing the hypothesis “ H smooth over S ” by “ H flat over S ”.

^{*}Cf. also J. P. Murre, *Representation of unramified functors. Applications*, Sémin. Bourbaki no. 294 (May 1965), th. 3 (p. 13).